Inverse functions

1 Graph $y = x$ on the same axes as the graphs of the two functions and look to see one of the graphs is a reflection of the other about $y = x$. If they are reflections, then they are inverses.

2

a No **b** Yes **c** No **d** Yes

e No **f** Yes **g** No

3

a

i $f^{-1}(x) = \frac{1}{8}(x + 9)$ **ii** $[-41, 7]$

iii $[-4, 2]$

b

i $f^{-1}(x) = \sqrt{x}$ **ii** $[0, \infty)$

iii $[0, \infty)$

c

i $f^{-1}(x) = \sqrt{16 - x^2}$ **ii** $[0, 4]$

iii $[0, 4]$

4

a Yes **b** No **c** Yes **d** Yes

e Yes **f** Yes **g** Yes **h** No

5

a Yes **b** No **c** Yes **d** Yes

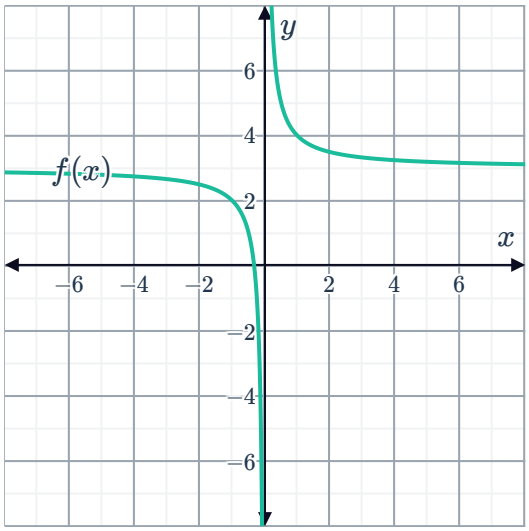
e No

Graphs of inverse functions

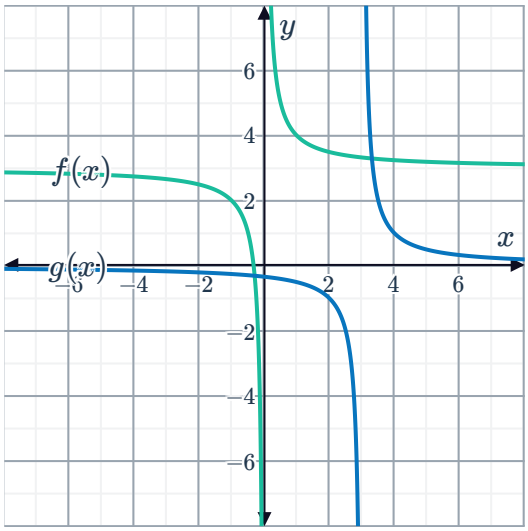
6

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a

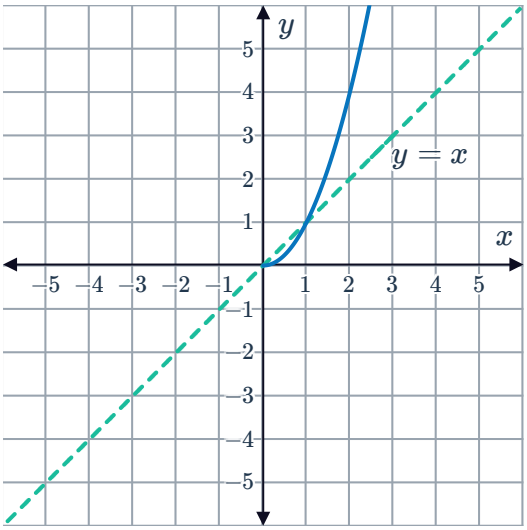


b



c Yes

7



8

a $x^3 - 2$

b $\sqrt[3]{x+2}$

c x

d x

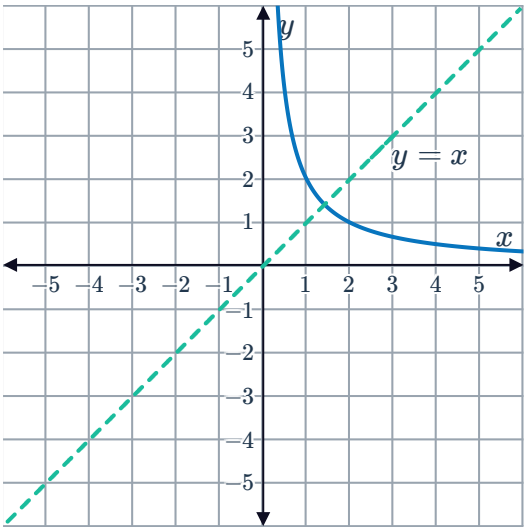
e

i Yes **ii** No **iii** Yes **iv** Yes

9

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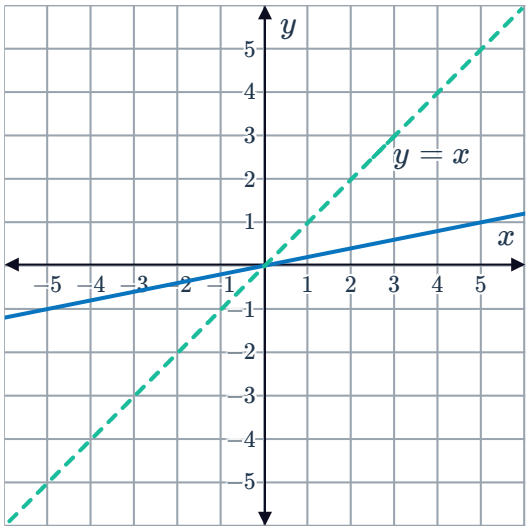
a



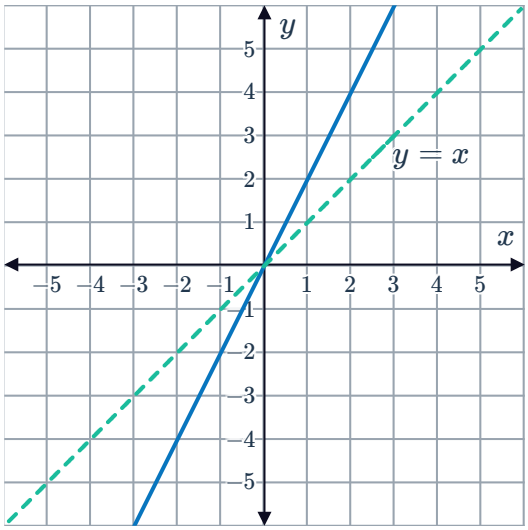
b The inverse graph is exactly the same as the original graph.

10

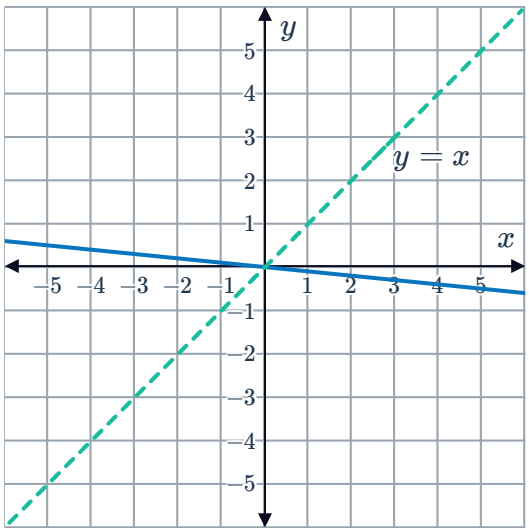
a



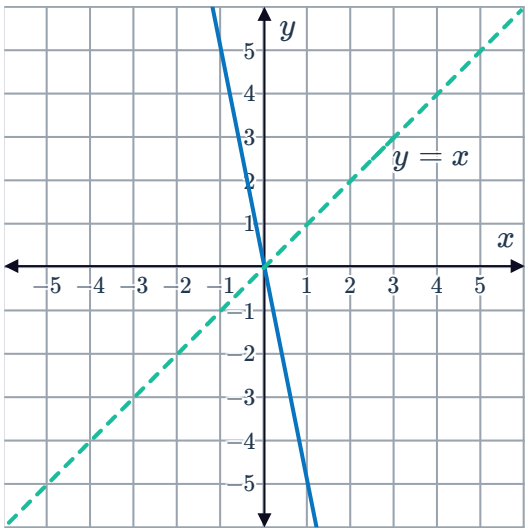
b



c

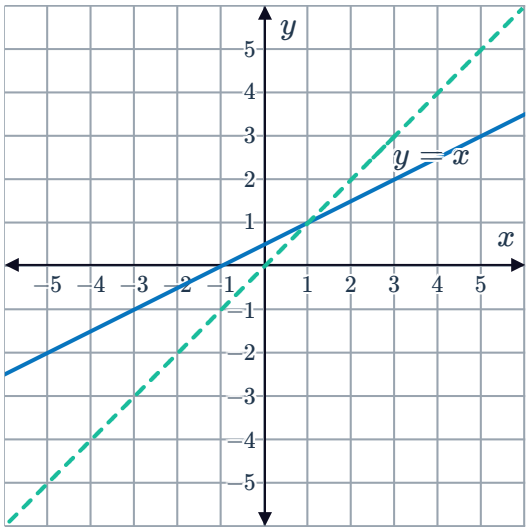


d

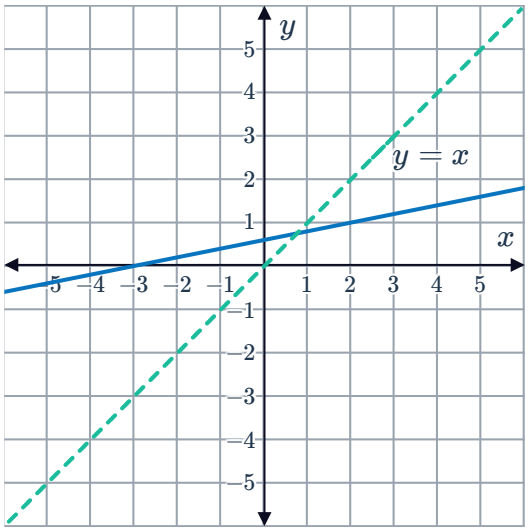


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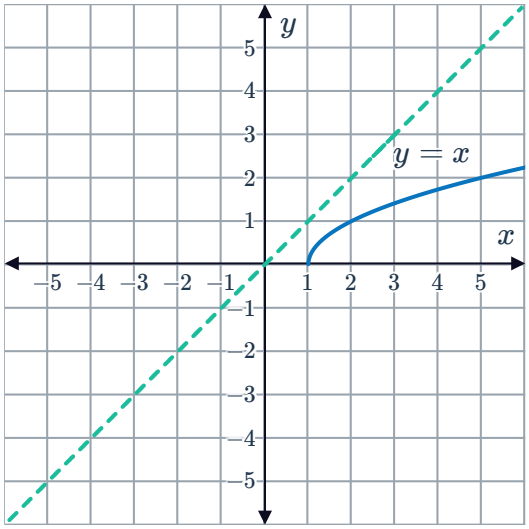
e



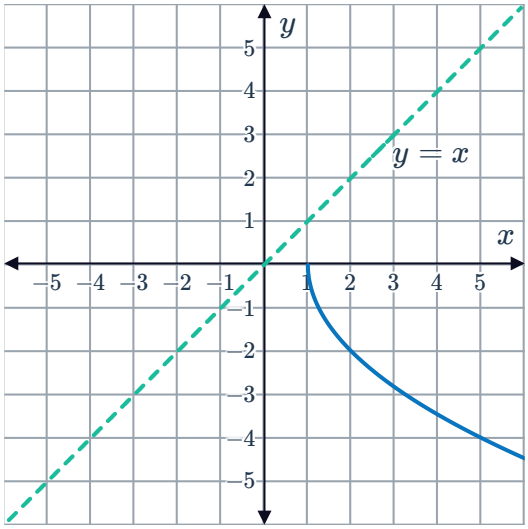
f



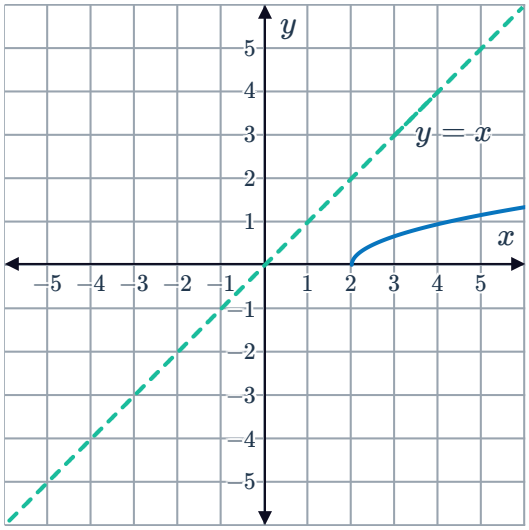
g



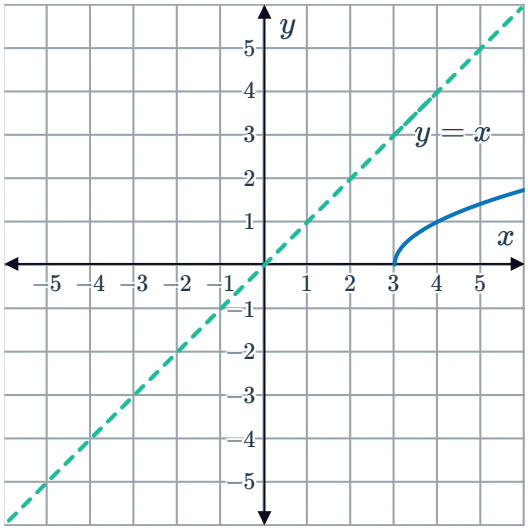
h



i



j

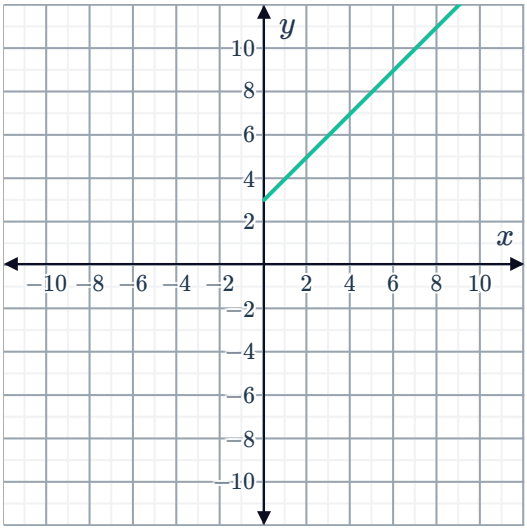


11

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a

i

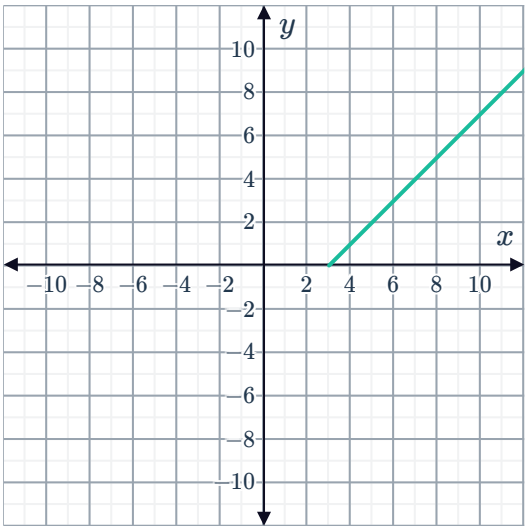


ii $f^{-1}(x) = x - 3$

iii $[3, \infty)$

iv $[0, \infty)$

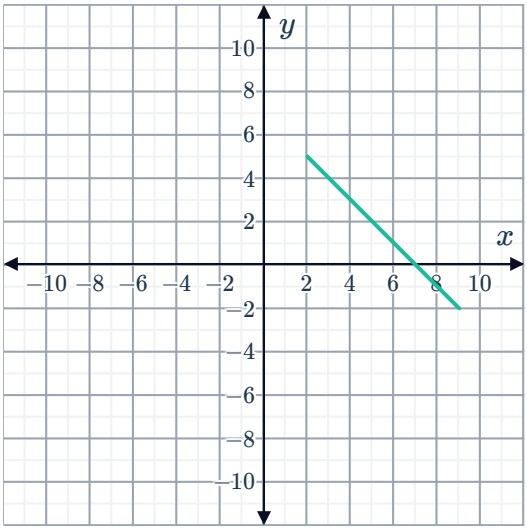
v



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b

i

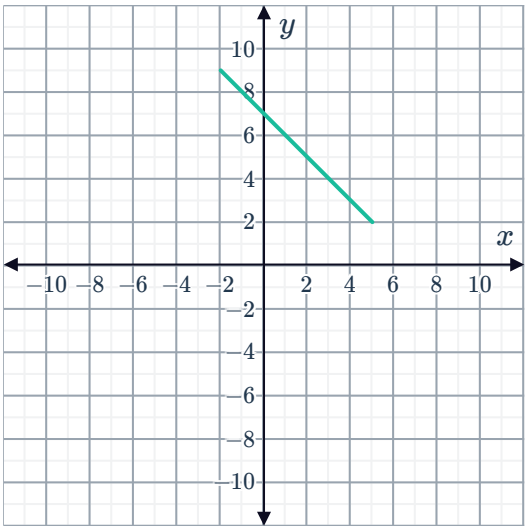


ii $f^{-1}(x) = 7 - x$

iii $[-2, 5]$

iv $[2, 9]$

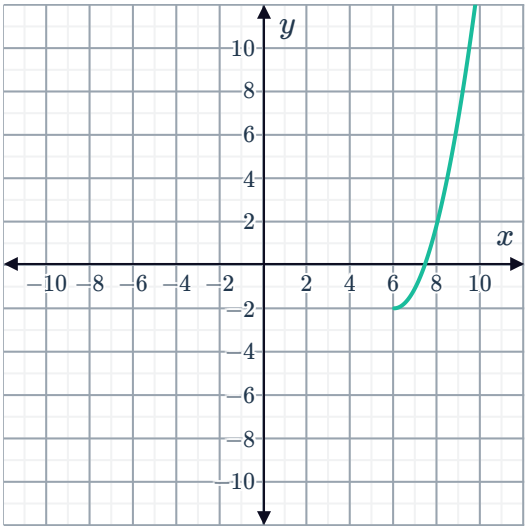
v



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c

i

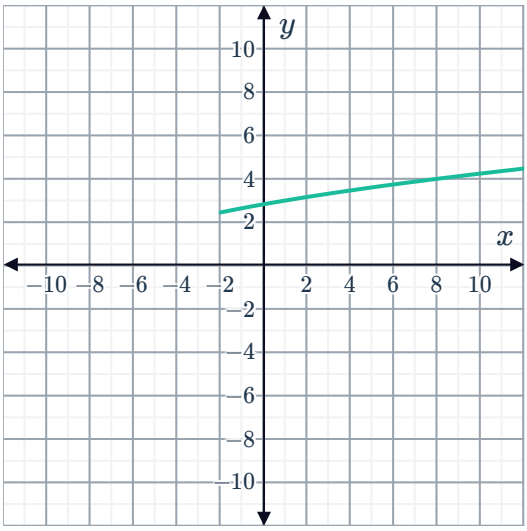


ii $f^{-1}(x) = \sqrt{x+2} + 6$

iii $[-2, \infty)$

iv $[6, \infty)$

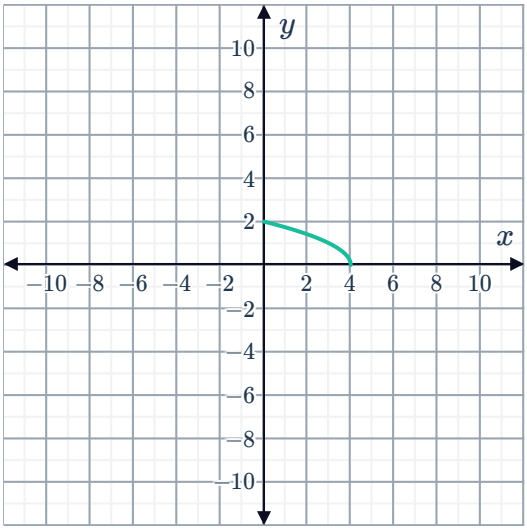
v



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d

i

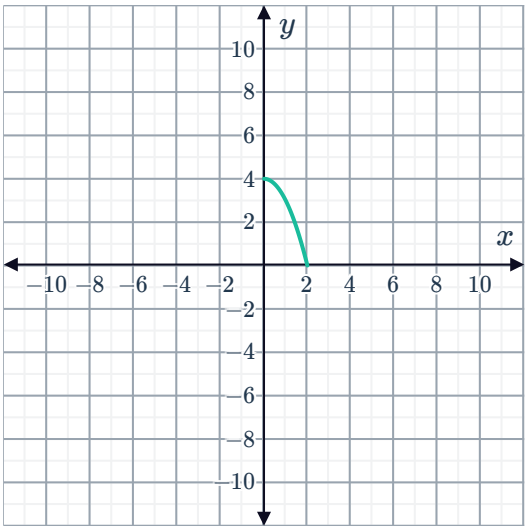


ii $f^{-1}(x) = 4 - x^2$

iii $(0, 2]$

iv $[0, 4)$

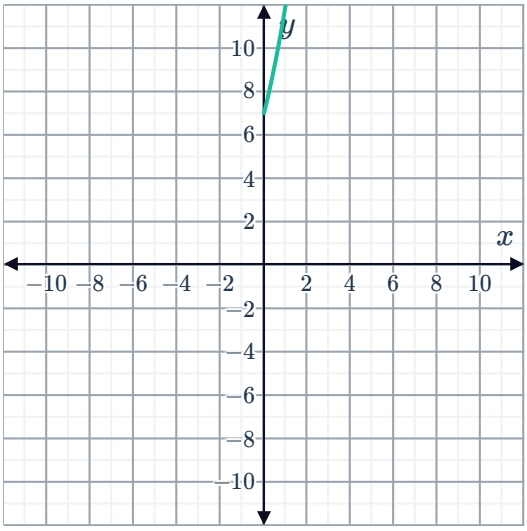
v



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e

i

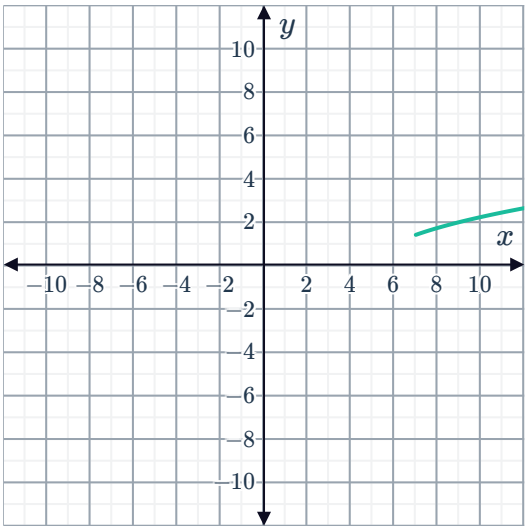


ii $f^{-1}(x) = \sqrt{x-3} - 2$

iii $[7, \infty)$

iv $[0, \infty)$

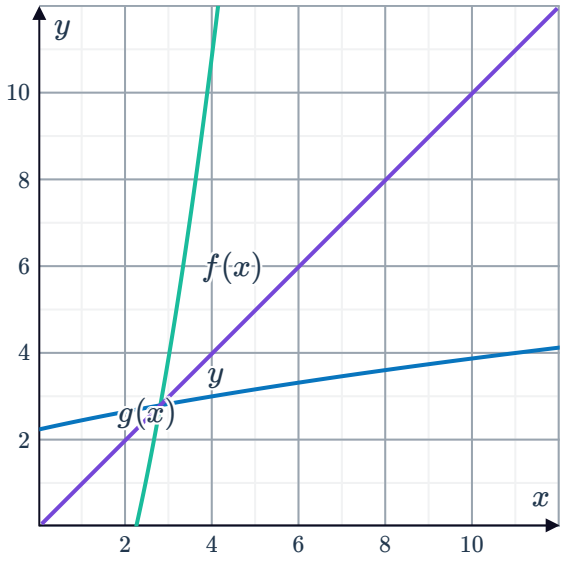
v



12

a $y = x$

b



c $f(x)$ is a reflection of $g(x)$ about $y = x$.

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Applications

13

a $d = 4.9t^2$

b $d = 122.5$

14

a -25°

b 30°

c $T^{-1}(x) = \frac{9}{5}x + 32$

d For converting Celsius temperatures x to Fahrenheit temperatures $T^{-1}(x)$

15

a Yes

b $t(d) = \sqrt{\frac{120 - d}{4.9}}$

c No solution provided

d No solution provided

e Yes

f 4 seconds

16

a

$$f(x) = \begin{cases} (x-2)^2 - 6, & x < 2 \\ (x-2)^2 - 6, & x \geq 2 \end{cases}$$

b

$$g(x) = \begin{cases} g(x), & x \leq -1 \\ g(x), & -1 < x < 6 \\ g(x), & x \geq 6 \end{cases}$$

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